

“Attention Is All You Need” — A Plain English Explainer

The paper that changed AI forever, explained for everyone

PAGE 1 — THE PROBLEM & THE BIG IDEA

First, Why Did This Paper Even Exist?

Before 2017, the best AI systems for reading and translating language worked kind of like humans reading one word at a time — slowly, left to right, always depending on what came before. These systems were called RNNs (Recurrent Neural Networks). They had two big problems: they were slow to train (because you couldn't do steps in parallel — each word had to wait for the previous one), and they were bad at remembering things from far back in a sentence. By the time an RNN reached the word “it” at the end of a long sentence, it had mostly “forgotten” what “it” referred to at the start.

The Big Idea — What Is “Attention”?

Imagine you're a translator and someone hands you this sentence: “The animal didn't cross the street because it was too tired.” What does “it” refer to — the animal or the street? You instantly look back at the whole sentence and figure it out: the animal. You didn't read left to right one word at a time — you jumped around and compared words to each other. That's exactly what “attention” does in AI.

Attention is a way for the model to look at every word in a sentence at the same time and decide: which other words matter most for understanding this word right now? The model learns to give each word a score — a weight — based on how relevant it is. Words that are highly relevant get a high score; words that aren't get a low score.

What the Paper Proposed

The 2017 paper by Vaswani et al. (Google Brain) had one bold claim: you don't need RNNs at all. Just use attention, over and over, in layers. That's literally the title — “Attention Is All You Need.” The model they built is called a Transformer. It became the foundation for every major AI language model since: GPT, BERT, Claude, ChatGPT — all Transformers.

The Transformer — A High-School Walkthrough

Think of the Transformer like a very focused study group. You give it a sentence (say, you want to translate “I love pizza” to French). It has two main parts: an Encoder (the “understander”) and a Decoder (the “writer”).

Step 1 — Turning Words into Numbers (Embeddings)

Computers can’t work with words directly, so first every word gets converted into a list of numbers — called an *embedding* — that captures meaning. Words with similar meanings get similar number patterns. “Cat” and “kitten” end up closer together in this number space than “cat” and “skyscraper.”

Step 2 — Telling the Model Where Words Are (Positional Encoding)

Here’s a trick: since a Transformer looks at all words at once (not one by one), it has no idea what order they’re in unless we tell it. So the authors added “positional encoding” — a little pattern of numbers added to each word’s embedding that signals its position in the sentence. Word 1 gets a slightly different signal than Word 2, and so on.

Step 3 — Multi-Head Self-Attention (The Core Magic)

This is the heart of the paper. For every word, the model asks three questions about every other word: (1) “What are you about?” — this creates a **Key**, (2) “What am I looking for?” — this creates a **Query**, and (3) “What information can you give me?” — this creates a **Value**. The model computes a score between each Query and each Key (a dot product, basically multiplication), softens those scores into probabilities with a softmax function, then uses those probabilities to create a weighted mix of all the Values. The result: each word now has a new, context-aware representation that “knows” about the whole sentence.

“Multi-head” just means this is done several times in parallel, with different learned weights each time. One “head” might learn to track grammar, another tracks meaning, another tracks references like “it” → “animal.” Each head sees the sentence through a different lens, and then all the results are combined.

Step 4 — Feed-Forward Layers and Stacking

After attention, each word’s new representation passes through a simple neural network (a feed-forward layer). This happens independently for each word. Then the whole attention + feed-forward block is repeated — stacked — 6 times in the original paper. Each layer refines the understanding deeper and deeper.

Why Was This Such a Big Deal?

Three reasons made the Transformer a revolution:

- **Speed** — Because attention looks at all words at once instead of one at a time, the whole process can be run in parallel on modern GPUs. Training that used to take weeks suddenly took days.
- **Long-range memory** — Attention doesn't "forget" what was at the start of the sentence. Every word can directly "look at" every other word regardless of distance. The signal doesn't have to travel through 50 steps like in an RNN.
- **It scaled** — The more data and compute you threw at it, the better it got. No other architecture before had scaled this cleanly.

The Results

On the standard machine translation benchmarks (translating English to German and English to French), the Transformer beat every previous model — and trained in a fraction of the time. It scored a BLEU score of 28.4 on English-to-German, a new record at the time. (BLEU is just a score that measures how close a translation is to a human one — higher is better.)

What Came After

The Transformer became the backbone of everything in modern AI:

- **BERT (2018, Google)** — Used just the Encoder part, trained to understand language for tasks like search and question answering. Google now uses BERT in its search engine.
- **GPT (2018–present, OpenAI)** — Used just the Decoder part, trained to predict and generate text. GPT-4 and ChatGPT are giant versions of this idea.
- **Claude (Anthropic)** — Also Transformer-based. The model you're talking to right now.
- **Vision Transformers (ViT)** — Transformers applied to images instead of words, now state of the art in computer vision too.

One-Line Summary

"Attention Is All You Need" said: stop reading one word at a time — instead, let every word look at every other word, score how relevant they are to each other, and build meaning from that. Simple idea, enormous impact.

A Quick Glossary

Term	What It Means
Embedding	Turning a word into a list of numbers that captures its meaning
Attention	A scoring system that figures out which words matter most for understanding each other
Query / Key / Value	The three ingredients of attention: what I'm looking for, what you are, what you can offer
Softmax	A math function that converts scores into probabilities that add up to 1
Multi-head attention	Running attention multiple times in parallel, each looking at different aspects
Encoder	The part that reads and understands the input sentence
Decoder	The part that generates the output sentence
BLEU score	A metric (0–100) measuring translation quality vs. human references

Paper reference: Vaswani, A. et al. (2017). "Attention Is All You Need." NeurIPS 2017. arXiv:1706.03762